

Standard calculation procedures and examples of calculation results for 4-quantization-level, 8-neighbor GLCMs. (By H.Nishiyama / Niigata Univ. 2026/06/27, Ver 2.1)

In PyRadiomics, the angle-related symmetrization parameter `weightingNorm` is set to 'None' by default. GLCMs are generated for each angle to calculate feature values, which are ultimately averaged.

<https://github.com/AIM-Harvard/pyradiomics>

https://github.com/auinen/pyradiomics_edu

It has been confirmed that this processing is likewise applicable to "Radiomics.jl," which runs on Julia.

<https://www.radiomicsjl.org/>

<https://github.com/pzaffino/Radiomics.jl>

Final Result
Average of 4 directions

Joint maximum	0.18112	0.15714	0.0240
Joint average	2.01818	2.01667	0.0015
Joint variance	0.96322	0.96401	-0.0008
Joint entropy	3.53613	3.60614	-0.0700

Results near GLCM-8
(simplified version) and
differences

8-neighborhood GLCM
(normalized)

0.119	0.143	0.062	0.031
0.143	0.157	0.045	0.048
0.062	0.045	0.010	0.017
0.031	0.048	0.017	0.024

